kader Academy

Machine Safety Specialist (CE)

Module 8: Ergonomics

TAB4: Risk reduction and standards
Tab4-8/9 20240819

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1

Ergonomics

Posture, strength and endurance

EN 1005-1: 2001 +A1: 2008

- Safety of machinery - Human physical strain - Part 1: Terms and definitions

EN 1005-2: 2003 +A1: 2008

- Safety of machinery - Human physical strain - Part 2: Manual handling of machinery and machine parts

EN 1005-3: 2002 +A1: 2008

- Safety of machinery - Human physical load - Part 3: Recommended maximum forces in machine work

EN 1005-4: 2005 +A1: 2008

- Safety of machinery - Human physical strain - Part 4: Evaluation of working postures and movements in machine work

EN 1005-5: 2007

- Safety of machinery - Human physical strain - Part 5: Risk assessment for repetitive high-frequency operations

2

Ergonomics

Posture, strength and endurance

In general:

- Prevent fatigue (physical and mental).
- Limit manual handling of heavy machinery or machine parts, other tools or automation.
- Enable operator to determine their own work pace.
- Avoid the need for lengthy repetitive actions.
- In general, maximum lifting mass (Mref) 25kg (women 18.75kg).
- Mass of 30, 35, 40 kg is allowed in exceptional cases, but only if it is not otherwise possible (state of the art); extra attention to education and training.

Ergonomics

Posture, strength and endurance

Example: A machine is designed in such a way that an operator has to take a product from a pallet and place it on a machine infeed belt.

<u>Principles:</u>

- 4 products are entered per minute (during an entire work shift, starting point is a combination of male and female operators)
- The product has a mass of 10kg and is easy to handle with sufficient grip
- Due to the size of the product, it must be held +/- 30cm in front of the body
- Height difference between the first layer of the pallet and the height of the infeed belt is 1m.

The question is whether this is ergonomically acceptable, so we are going to determine this using the EN 1005-2 standard.

Tab4-8/9-Sh.5

Ergonomics

Posture, strength and endurance

- The standard has 3 methods to determine this. In this example, method 2 (calculation using tables).

  The other 2 methods consist of a scan based on preconditions and a complete calculation.

- The formula we need:
  $R_{ML2} = M_{ref} \times V_M \times D_M \times H_M \times A_M \times C_M \times F_M$,

  $M_{ref} = 25*0{,}85 = 21{,}25$kg

Table 1 — Reference mass (M_{ref}) taking into consideration the intended user population

Field of application	M_{ref} [kg]	Percentage of F and M	F_{emales}	M_{ales}	Population group	
Domestic use[a]	5	Data not available			Children and the elderly	Total population
	10	99	99	99	General domestic population	
Professional use (general)[b]	15	95	90	99	General working population, including the young and old	General working population
	25	85	70	90	Adult working population	
Professional use (exceptional)[c]	30 35 40	Data not available			Special working population	Special working population

[a] When designing a machine for domestic use, 10 kg should be used as a general reference mass in the risk assessment. If children and elderly are included in the intended user population, the reference mass should be lowered to 5 kg.

[b] When designing a machine for professional use, a reference mass of 25 kg should not be exceeded in general.

[c] While every effort should be made to avoid manual handling activities or reduce the risks to the lowest possible level, there may be exceptional circumstances where the reference mass might exceed 25 kg (e.g. where technological developments or interventions are not sufficiently advanced). Under these special conditions other measures have to be taken to control the risk according to EN 614-1 (e.g. technical aids, instructions and / or special training for the intended operator group).

Tab4-8/9-Sh.6

Ergonomics

Posture, strength and endurance

$R_{ML2} = M_{ref} \times V_M \times D_M \times H_M \times A_M \times C_M \times F_M$,

$R_{ML2} = 21{,}25 \times 0{,}93 \times 1 \times 0{,}83 \times 1 \times 1 \times 0{,}45$

$R_{ML2} = 7{,}4$kg

If all criteria are met, then determine the level of risk by calculating the recommended mass limit (R_{ML2}) using the multipliers provided in Table 2 and then the risk index (R_i) as follows:

$$R_i = \frac{\text{actual mass}}{R_{ML2}}$$

4.3.3.2.3 Select the action required (step 3)

— If $R_i \leq 0{,}85$, the risk may be regarded as tolerable (green).

— $0{,}85 < R_i < 1{,}0$ indicates that significant risk exists (yellow). It is recommended that:

  — method 3 is applied in order to identify how the risk may be reduced or

  — the machinery be either redesigned or

  — ensure that the risk is tolerable.

$R_i \geq 1{,}0$ means that redesign is necessary. The design can be improved by changing the situations that lead to low multipliers (red).

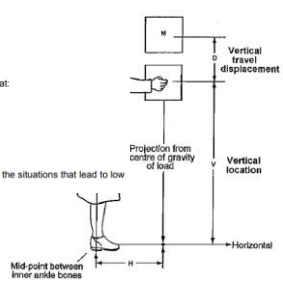


Table 2 — Multipliers for the calculation of the recommended mass limit (R_{ML2})

Vertical multiplier (V_M)							
Vertical location[3] [cm]	0	25	50	75	100	130	> 175
Multiplier	0,78	0,85	0,93	1,00	0,93	0,84	0,00
Distance multiplier (D_M)							
Vertical displacement[3] [cm]	25	30	40	50	70	100	> 175
Multiplier	1,00	0,97	0,93	0,91	0,88	0,87	0,00
Horizontal multiplier (H_M)							
Horizontal location[3] [cm]	25	30	40	50	56	60	> 63
Multiplier	1,00	0,83	0,63	0,50	0,45	0,42	0,00
Asymmetric multiplier (A_M)							
Angle of asymmetry[3] [°]	0	30	60	90	120	135	> 135
Multiplier	1,00	0,90	0,81	0,71	0,62	0,57	0,00

Coupling multiplier (C_M)			
Quality of grip	Good	Fair	Poor
Description	see annex C	see annex C	see annex C
Multiplier	1,00	0,95	0,90

Frequency multiplier (F_M)[4]

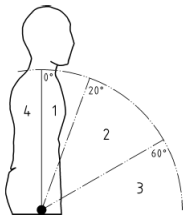
	Frequency						
Hz	0,0033	0,0166	0,0666	0,1000	0,1500	0,2000	> 0,2500
lifts/min	0,20	1	4	6	9	12	> 15
Duration (d) d ≤ 1h	1,00	0,94	0,84	0,75	0,52	0,37	0,00
1h < d ≤ 2h	0,95	0,88	0,72	0,50	0,30	0,00	0,00
2h < d ≤ 8h	0,85	0,75	0,45	0,27	0,00	0,00	0,00

Tab4-8/9-Sh.7

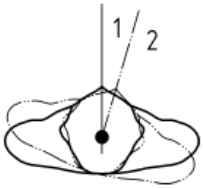
Ergonomics

Posture, strength and endurance

- Naast een maximale te hanteren massa in het ontwerp van een machine worden in EN 1005-4 o.a. ook eisen gesteld aan houding en bewegingen van de romp zoals i.r.t. de frequentie en tijdsduur:



Stoop



Turn

Lateral bends

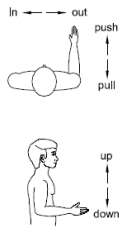
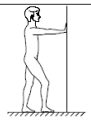
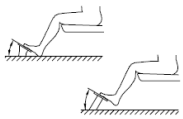
Tab4-8/9-Sh.8

Ergonomics

Posture, strength and endurance

- For the purpose of operating or handling machine parts by hand or foot, maximum forces have been defined in relation to:
  - manual work
  - work from the entire arm
  - work for which the body must be used
  - work with the foot

- Again, this is in contrast to frequency and duration.

Activity		Professional use F_B in N	Domestic use F_B in N
	Hand work (one hand):		
	Power grip	250	184
	Arm work (sitting posture, one arm):		
	- upwards	50	31
	- downwards	75	44
	- outwards	55	31
	- inwards	75	49
	- pushing		
	- with trunk support	275	186
	- without trunk support	62	30
	- pulling		
	- with trunk support	225	169
	- without trunk support	55	28
	Whole body work (standing posture):		
	- pushing	200	119
	- pulling	145	96
	Pedal work (sitting posture, with trunk support):		
	- ankle action	250	154
	- leg action	475	308

Tab4-8/9-Sh.9

Ergonomics

Human-machine interface (signaling, marking and operation)

Nederlandse norm

NEN-EN-IEC 61310-1

(en; fr)

Safety of machinery - Indication, marking and actuation - Part 1: Requirements for visual, accoustic and tactile signals (IEC 61310-1:2007,IDT)

Nederlandse norm

NEN-EN-IEC 61310-3

(en; fr)

Safety of machinery - Indication, marking and actuation - Part 3: Requirements for the location and operation of actuators (IEC 61310-3:2007,IDT)

Nederlandse norm

NEN-EN-IEC 61310-2

(en; fr)

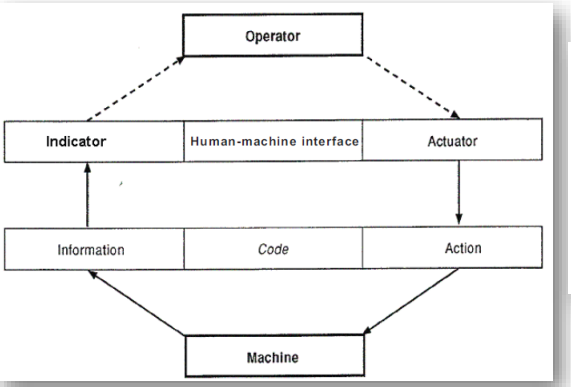
Safety of machinery - Indication, marking and actuation - Part 2: Requirements for marking (IEC 61310-2:2007,IDT)

Tab4-8/9-Sh.10

Ergonomics

Human-machine interface (signaling, marking and operation)

- Clear, unambiguous interaction between operator and machine
- Clear information (displays, texts, pictograms))



EN IEC 61310-1 Figure 1

Table 2 – Meaning of colours – General principles

Colour	Meaning		
	Safety of persons or environment	Condition of process	State of equipment
RED	Danger	Emergency	Faulty
YELLOW	Warning/caution	Abnormal	Abnormal
GREEN	Safe	Normal	Normal
BLUE	Mandatory significance		
WHITE GREY BLACK	No specific meaning assigned		

IEC 60073 Table 2

Tab4-8/9-Sh.11

Colour of control elements (implementation of standard colours in EN IEC 60204-1)

Control element	Colour		
	preference	alternative	Not allowed
Start / On	**white**	Grey Black green	red
Stop / Off	**black**	White Grey Red (not near emergency stop / emergency)	green
Combined start/stop	**White** **Black** **grey**		Red Yellow green
Hold-to-run	**White** **Black** **grey**		Red Yellow green
Reset	**blue**	White Grey black	green
Combined reset/stop	**black**	White grey	green
Emergency stop / Emergency off	**Red with yellow background**	-	All other colour combinations

Tab4-8/9-Sh.12

Colour of control elements (implementation of standard colours in EN IEC 60204-1)

Table 4 – Colours for indicator lights and their meanings with respect to the condition of the machine

Colour	Meaning	Explanation	Action by operator
RED	Emergency	Hazardous condition	Immediate action to deal with hazardous condition (for example switching off the machine supply, being alert to the hazardous condition and staying clear of the machine)
YELLOW	Abnormal	Abnormal condition Impending critical condition	Monitoring and/or intervention (for example by re-establishing the intended function)
BLUE	Mandatory	Indication of a condition that requires action by the operator	Mandatory action
GREEN	Normal	Normal condition	Optional
WHITE	Neutral	Other conditions; may be used whenever doubt exists about the application of RED, YELLOW, GREEN, BLUE	Monitoring

Indicating towers on machines should have the applicable colours in the following order from the top down; RED, YELLOW, BLUE, GREEN and WHITE.

Control elements

- Prevent nuisance from personal protective equipment
- Visible and recognizable
- Colors and marking
- Secure placement
- Logical effect
- Fixed layout
- Tax
- Unequivocal
- Ergonomic
- View of all zones
- Signalling and indication

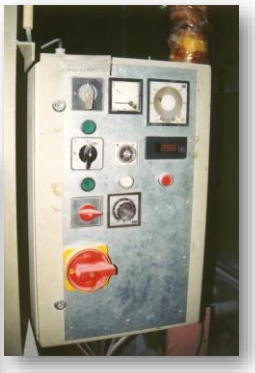
Table 2 – Classification of actions

Nature of actuator		Nature of action		Direction of action	
				Group 1	Group 2
Handwheel, handle, knob, etc.		Rotation		Clockwise	Anticlockwise
Grip, lever, push-pull button, etc., with essentially linear motion[a]		Vertical motion		Upwards	Downwards
		Horizontal motion	Right-left	To the right	To the left
			Forward-backward[a]	Away from the operator	Towards the operator
Nature of actuator set		**Nature of action**		**Point of application of action**	
				Group 1	**Group 2**
Set of grips, push-buttons, rods, pull cords, etc., with opposite effects	One above the other	Pressure, traction, etc.		Action on upper device	Action on lower device
	One beside the other			Action on right device	Action on left device
Nature of actuator set		**Nature of action**		**Classification of action**	
VDT with XY-VDU Controller		Movement and actuation (click)		Direction of action and point of application: not classified[b]	
Keyboard		Typing of keys			
Sensitive area		Touching			

[a] Further information is given in IEC 60447.

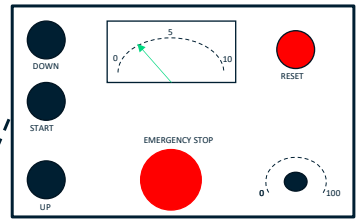
[b] As far as possible, the rules of the upper part of Table 2 should be applied.

EN IEC 61310-3 Table 2



Control elements

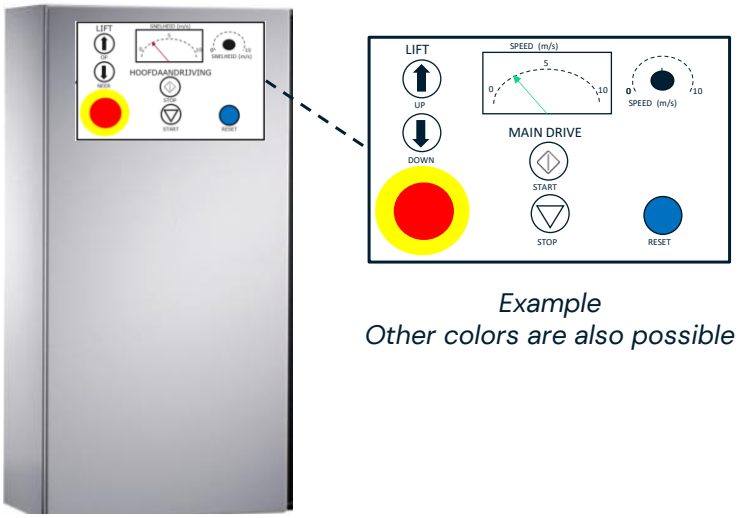
Human-machine interface (signaling, marking and operation)



What could be better?

Control elements

Human-machine interface (signaling, marking and operation)



Example
Other colors are also possible

Ergonomics

Human-machine interface (signaling, marking and operation)

EN IEC 61310-1 & EN ISO 7010

Prohibition — Red = danger to life

Mandatory — Blue = mandatory action

Warning — Yellow = Caution / Abnormal and Warning

Information — Green = Safe / normal operation

Ergonomics

Human-machine interface (signaling, marking and operation)

EN IEC 61310-1 & EN ISO 7010, Examples

Ergonomics

Biomechanical limits

TECHNICAL SPECIFICATION

ISO/TS 15066

First edition
2016-02-15

Robots and robotic devices — Collaborative robots

Robots et dispositifs robotiques — Robots coopératifs

Tab4-8/9-Sh.19

Ergonomics
Biomechanical limits

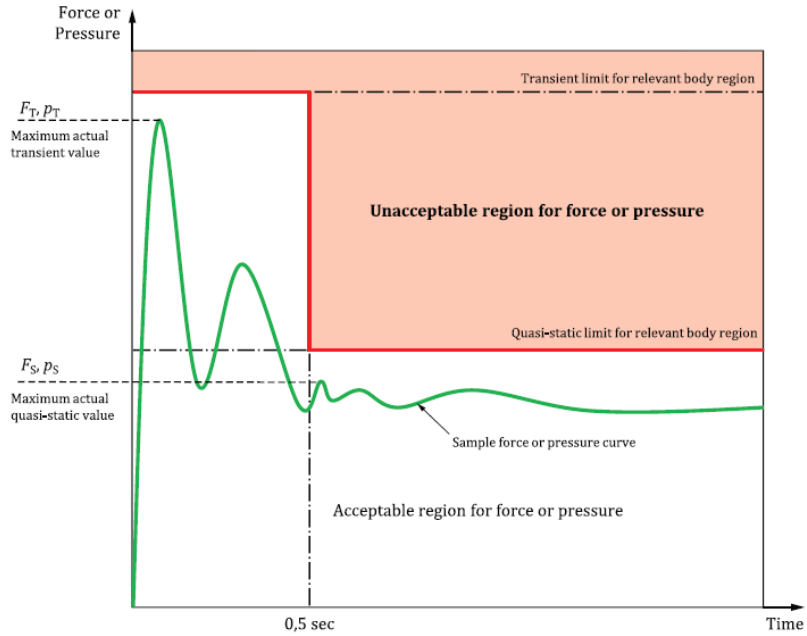


Figure 4 — Graphical representation of acceptable and unacceptable forces or pressures

Ergonomics
Biomechanical limits

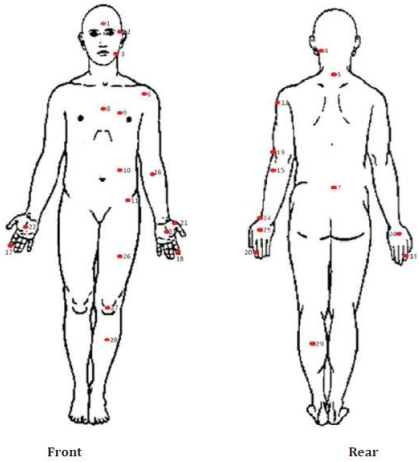


Figure A.1 — Body model

Table A.1 — Body model descriptions

Body region	Specific body area		Front/Rear
Skull and forehead	1	Middle of forehead	Front
	2	Temple	Front
Face	3	Masticatory muscle	Front
Neck	4	Neck muscle	Rear
	5	Seventh neck vertebra	Rear
Back and shoulders	6	Shoulder joint	Front
	7	Fifth lumbar vertebra	Rear
Chest	8	Sternum	Front
	9	Pectoral muscle	Front
Abdomen	10	Abdominal muscle	Front
Pelvis	11	Pelvic bone	Front
Upper arms and elbow joints	12	Deltoid muscle	Rear
	13	Humerus	Rear
Lower arms and wrist joints	14	Radial bone	Rear
	15	Forearm muscle	Rear
	16	Arm nerve	Front
Hands and fingers	17	Forefinger pad D [a]	Front
	18	Forefinger pad ND [a]	Front
	19	Forefinger end joint D [a]	Rear
	20	Forefinger end joint ND [a]	Rear
	21	Thenar eminence	Front
	22	Palm D [a]	Front
	23	Palm ND [a]	Front
	24	Back of the hand D [a]	Rear
	25	Back of the hand ND [a]	Rear
Thighs and knees	26	Thigh muscle	Front
	27	Kneecap	Front
Lower legs	28	Middle of shin	Front
	29	Calf muscle	Rear

[a] D = dominant body side; ND = non-dominant body side.

Ergonomics

Biomechanical limits

Table A.2 — Biomechanical limits

Body region		Specific body area	Quasi-static contact		Transient contact	
			Maximum permissible pressure [a] p_s N/cm²	Maximum permissible force [b] N	Maximum permissible pressure multiplier [c] P_T	Maximum permissible force multiplier [c] F_T
Skull and forehead [d]	*1*	*Middle of forehead*	*130*	*130*	*not applicable*	*not applicable*
	2	*Temple*	*110*		*not applicable*	
Face [d]	*3*	*Masticatory muscle*	*110*	*65*	*not applicable*	*not applicable*
Neck	4	Neck muscle	140	150	2	2
	5	Seventh neck muscle	210		2	
Back and shoulders	6	Shoulder joint	160	210	2	2
	7	Fifth lumbar vertebra	210		2	2
Chest	8	Sternum	120	140	2	2
	9	Pectoral muscle	170		2	
Abdomen	10	Abdominal muscle	140	110	2	2
Pelvis	11	Pelvic bone	210	180	2	2
Upper arms and elbow joints	12	Deltoid muscle	190	150	2	2
	13	Humerus	220		2	
Lower arms and wrist joints	14	Radial bone	190	160	2	2
	15	Forearm muscle	180		2	
	16	Arm nerve	180		2	

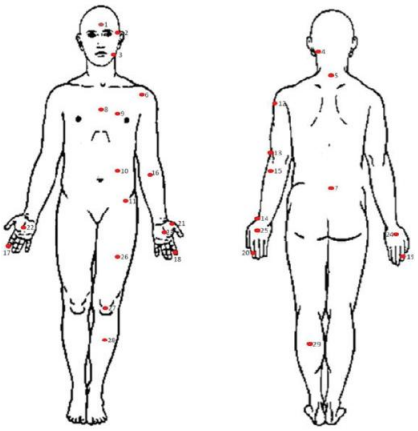


Figure A.1 — Body model

Ergonomics

Biomechanical limits

Table A.2 *(continued)*

Body region		Specific body area	Quasi-static contact		Transient contact	
			Maximum permissible pressure [a] p_s N/cm²	Maximum permissible force [b] N	Maximum permissible pressure multiplier [c] P_T	Maximum permissible force multiplier [c] F_T
Hands and fingers	17	Forefinger pad D	300	140	2	2
	18	Forefinger pad ND	270		2	
	19	Forefinger end joint D	280		2	
	20	Forefinger end joint ND	220		2	
	21	Thenar eminence	200		2	
	22	Palm D	260		2	
	23	Palm ND	260		2	
	24	Back of the hand D	200		2	
	25	Back of the hand ND	190		2	
Thighs and knees	26	Thigh muscle	250	220	2	2
	27	Kneecap	220		2	
Lower legs	28	Middle of shin	220	130	2	2
	29	Calf muscle	210		2	

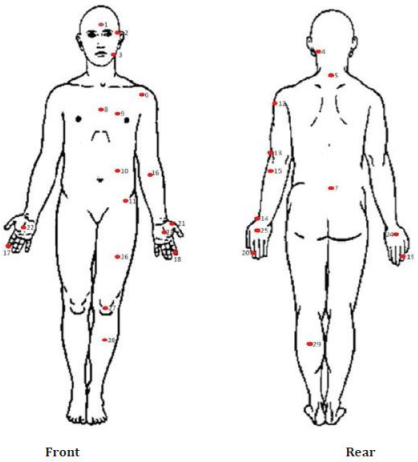
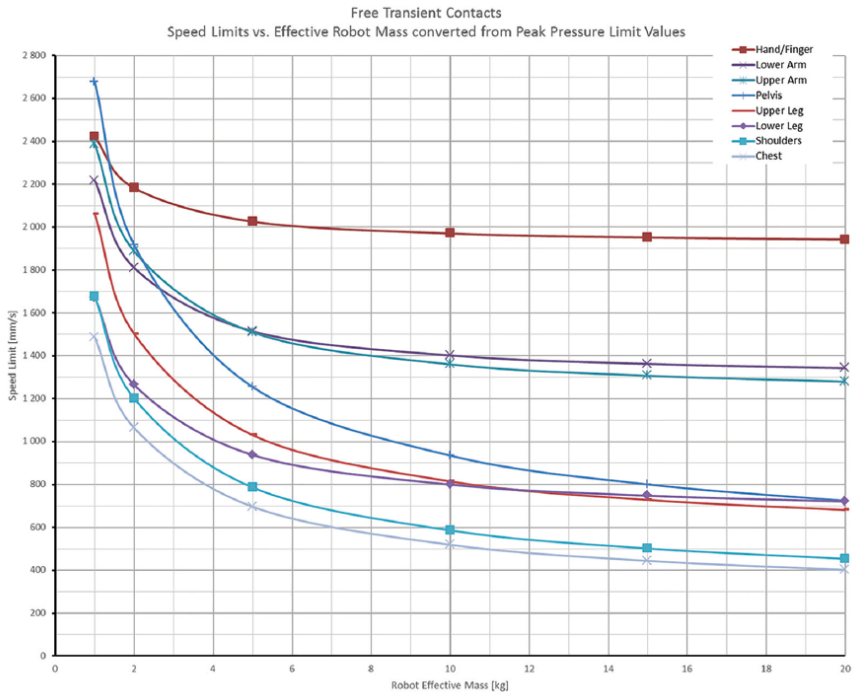


Figure A.1 — Body model

Ergonomics
Biomechanical limits